NE763 Exam 2

Frederick Humm
7 Oct 2026

97

1) SFRs have inherent safety built in via the following phenomena:
- There is a large temperature range between the coolant's melting point and boiling point (~700C), allowing for significant margin during transient power conditions.
- The liquid metal coolant (Na, Pb) is able to operate at near-atmospheric temperatures, with a boiling point of 98C at atmosphere.
- The high heat capacity of the fuel creates a long thermal response time. This causes an innate negative reactivity feedback loop with transient increasing temperatures.
- The metal fuel is able to maintain its structure even in the case of cladding damage/rupture for long periods of time.

6/6

2) Alloying uranium, specifically with Zr, helps reduce overall swelling due to fission gases and reduce FCMI and therefore damage to the cladding. Alloying does this by providing point defect sinks. Alloying also ~~increases~~ broadens the temperature range of the γ-phase and therefore also increases the overall melting temperature of the fuel.

6/6

3) Metallic fuel systems need a low smear density (~75%) to account for FCMI. Early designs of metallic fuel reactors did not account for the swelling of fuel due to non-soluble fission product production (e.g., rare earth/lanthanides). As the fuel swelled, it contacted the cladding, adding mechanical stress and causing cladding rupture. A low smear density allows the fuel to swell until fission gases are released. The large plenum is required in order to allow for this large amount of fission gas release (acts as a "vent" for the gases). – why? cladding stresses

10/10

4) Constituent redistribution in metallic fuels is the process of Zr diffusing up the temperature gradient while U diffuses down, creating 3 distinct regimes that follow phase-driven isotherms (roughly): 1) Zr-enriched center, 2) Zr-depleted intermediary zone, 3) an as-fabricated Zr concentration outer zone near the cladding. Constituent redistribution is both a temperature driven process (and therefore varies axially in the fuel slug) and chemical potential-driven process.

Constituent redistribution can cause lower fission rates and centerline temperatures in U-Zr fuels, but in U-Pu-Zr fuels a higher centerline temperature and lower inner melting point can occur due to the diffusion of Pu up the temperature gradient as well. Redistribution can also cause more anisotropic swelling and worsening the effects of fuel swelling.

14/14

5) Thermal conductivity initially worsens with burnup in metallic fuels due to the formation of pores. As porosity increases, the pores have low thermal conductivity, causing a net degredation of thermal conductivity. However, eventually a high enough burnup will cause the pores to interconnect and fill with bond sodium, which has a high thermal conductivity. As a result, thermal conductivity only degrades due to initial burnup before being counteracted by the inclusion of bond sodium.

8/8

6) FCCI is Fuel-Cladding Chemical Interaction and the driving force of fuel performance in metallic fuels. FCCI is driven by the diffusion of fuel, cladding, and fission product species at the boundary of the fuel and cladding. FCCI is typically measured as FCCI "layer depth".

There are two primary adverse effects of FCCI:
1) Forming localized low melting point regions in the fuel
2) Weakening the structural integrity of the cladding.

These adverse effects are driven by various fuel/FP/cladding species diffusing across the boundary and infiltrating the composition of the receptive layer. The primary fuel species involved in FCCI are U, Pu, and Zr. The inclusion of Zr actually limits FCCI, as well as creating a localize, protective "rind" around the fuel. Lanthanide fission products are the most involved FPs, – Ce + Nd with higher temperatures and burnup increasing the impact of FCCI due to larger amounts of fission products diffusing to the outer layer. In the cladding, Fe, Ni, and Cr play the largest role in FCCI, with Fe at a high risk of diffusing into the fuel.

14/14

7) The restructuring of MOX fuel creates the following four layers inside the fuel pin:
1) central void
2) columnar region (extended grains)
3) equiaxed region (larger grains than as-fabricated)
4) as-fabricated/as-sintered region

This restructuring occurs at very high temperatures due to the migration of pores up the temperature gradient. Due to the low thermal conductivity inside the pore, there is a large temperature difference between the inner and outer surfaces of the pore. This allows material to evaporate off the inner wall and condense onto the outer wall, effectively diffusing the pore up the temperature gradient until the pores coalesce in the central void. The >1800C region forms the long, extended grain structures seen in the columnar region. – why the columns? Why only grain growth below?

11/12

8) MOX restructuring also changes the composition of the fuel. UO_x is much more volatile than PuO_x, therefore the UO_2 is more likely to diffuse down the temperature gradient via the process described above. This forms a Pu-rich inner layer in the fuel, which both increases the inner fission rate and decreases the melting point of the inner surface. Areas of high Pu concentration can also form due to poor mixing in the manufacturing process, creating Pu agglomerates.

MOX fuel is initially fabricated with a ~~high~~ low stoichiometric ratio, with an O/M just below 2.00, 1.93-1.96 Oxygen then diffuses down the temperature gradient of the fuel, bringing O/M close to 2.00 at the outer edges. This ratio increases until Mo fission product begins to oxidize, keeping O/M at about 2.00.

9/10

9) JOG is Joint Oxide Gain, where an oxide layer can build up between the fuel and the cladding due to the diffusion of volatile fission products down the temperature gradient. Compounds (with Cs/Mo/O primarily) form in low temperature areas. JOG has two primary impacts:
1) Improving the overall thermal conductivity of the fuel because oxides have higher thermal conductivity than a gap.
2) Providing a "buffer" between the fuel and cladding and reducing the impacts of swelling.

– promotes FCCI

11/12

10) Corrosion due to a liquid sodium coolant is a significant concern in SFR structural material selection. Sodium corrosion can occur through one of the two following methods:
1) Dissolving alloy constituents into the sodium (dependent on factors like alloy solubility in sodium, coolant temperature, coolant velocity).
2) Alloy constituents reacting with impurities in the sodium coolant (driving force for the need to test for and remove impurities in the coolant).

8/8

Notably, interaction between the sodium coolant and (primarily stainless steel) structural elements does not cause corrosion. An effect of sodium corrosion in SFRs is the net mass transfer due to alloy constituents dissolving into the sodium in high-temperature areas of the coolant loop and then depositing onto low-temperature areas of the loop.